# Final Project

The goal is to prepare a database for the university with tables like subjects, students...

```
CREATE TABLE subject_types (
  type_id NUMBER PRIMARY KEY,
  type_name VARCHAR2(40)
);

CREATE TABLE subjects (
  subject_id NUMBER PRIMARY KEY,
  subject_name VARCHAR2(80)
);

CREATE TABLE subject_subject_types (
  subject_id NUMBER REFERENCES subjects(subject_id),
  type_id NUMBER REFERENCES subject_types(type_id),
  PRIMARY KEY (subject_id, type_id)
);

CREATE TABLE students (
  student_id NUMBER PRIMARY KEY,
  first_name VARCHAR2(40),
  last_name VARCHAR2(40)
);

CREATE TABLE teachers (
  teacher_id NUMBER PRIMARY KEY,
  first_name VARCHAR2(40),
  last_name VARCHAR2(40)
);

CREATE TABLE teacher_salary (
  teacher_id NUMBER REFERENCES teachers(teacher_id),
  salary NUMBER,
  employment_type VARCHAR2(40)
);

CREATE TABLE grades (
  grade_id NUMBER PRIMARY KEY,
  student_id NUMBER REFERENCES students(student_id),
  subject_id NUMBER REFERENCES subjects(subject_id),
  type_id NUMBER REFERENCES subject_types(type_id),
  grade NUMBER
);
```

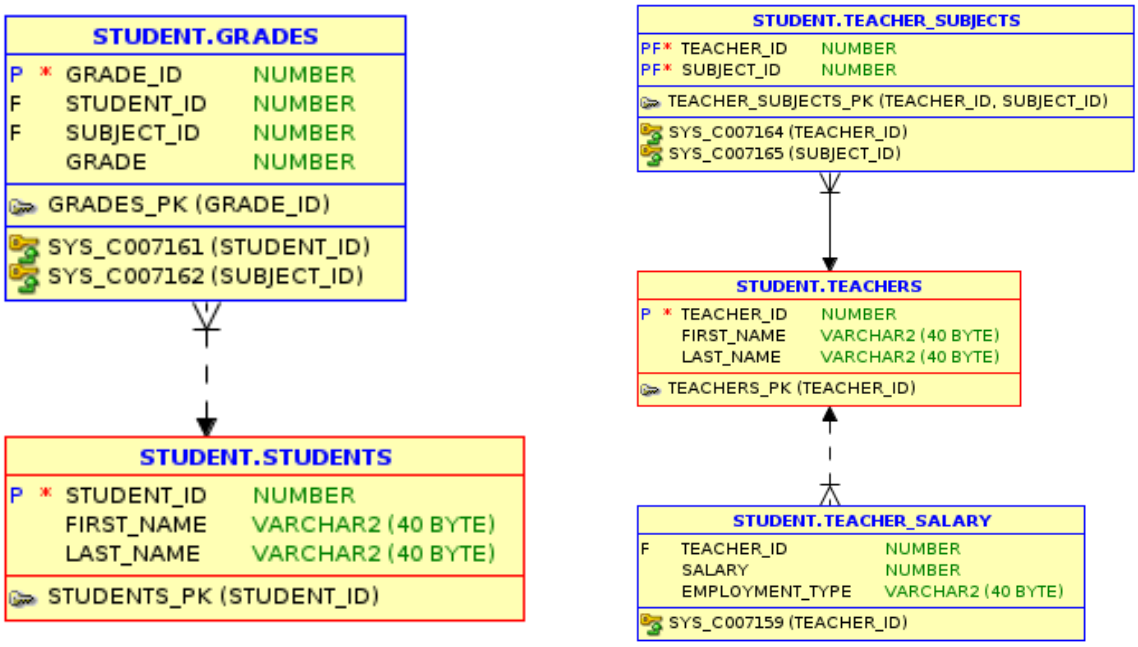
```
CREATE TABLE teacher_subjects (
  teacher_id NUMBER REFERENCES teachers(teacher_id),
  subject_id NUMBER REFERENCES subjects(subject_id),
  PRIMARY KEY (teacher_id, subject_id)
);
```

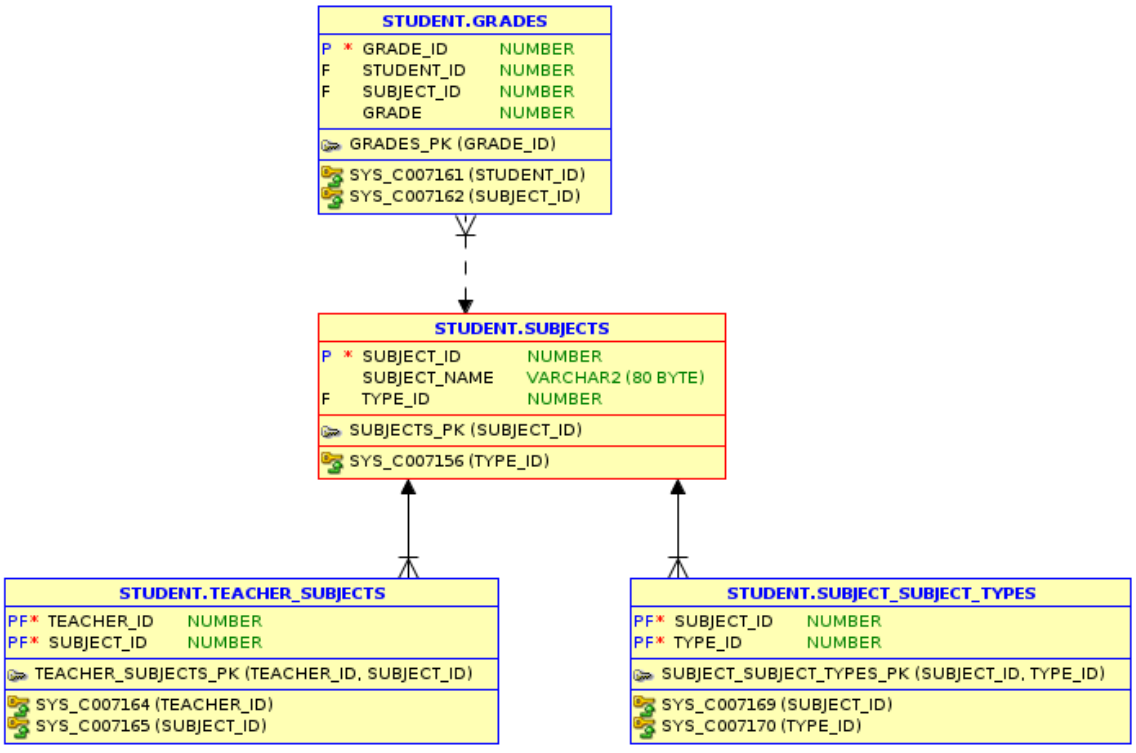
ER diagram shows the relationships between tables in the university database. The database consists of the following tables:

- SUBJECT_TYPES – stores different types of subjects, such as Lecture, Exercise, and Laboratory
- SUBJECTS – stores information about university subjects
- SUBJECT_SUBJECT_TYPES – stores the relationship between subjects and subject types, because one subject can have more than one type
- STUDENTS – stores student information
- TEACHERS – stores teacher information
- TEACHER_SALARY – stores teacher salaries and employment types
- GRADES – stores students' grades for subjects and subject types
- TEACHER_SUBJECTS – stores the relationship between teachers and subjects

Relationships:
• One subject can have many subject types
• One subject type can belong to many subjects
• One student can receive many grades
• One subject can have many grades
• One subject type can be connected with many grades
• One teacher can teach many subjects
• One subject can be taught by many teachers
• One teacher has salary and employment information stored in the TEACHER_SALARY table

ER diagrams for university database:



Now that tables are created, we need to add data:

Three different types of subjects:
```
INSERT INTO subject_types VALUES (1, 'Lecture');
INSERT INTO subject_types VALUES (2, 'Exercise');
INSERT INTO subject_types VALUES (3, 'Laboratory');
```

Six different subjects:
```
INSERT INTO subjects VALUES (1, 'Databases');
INSERT INTO subjects VALUES (2, 'Mathematics');
INSERT INTO subjects VALUES (3, 'Programming');
INSERT INTO subjects VALUES (4, 'Computer Networks');
INSERT INTO subjects VALUES (5, 'Operating Systems');
INSERT INTO subjects VALUES (6, 'English');
```

Linking subject with it's types:
```
-- Databases: Lecture + Laboratory
INSERT INTO subject_subject_types VALUES (1, 1);
INSERT INTO subject_subject_types VALUES (1, 3);

-- Mathematics: Lecture + Exercise
INSERT INTO subject_subject_types VALUES (2, 1);
INSERT INTO subject_subject_types VALUES (2, 2);

-- Programming: Lecture + Laboratory
INSERT INTO subject_subject_types VALUES (3, 1);
INSERT INTO subject_subject_types VALUES (3, 3);

-- Computer Networks: Lecture + Laboratory
INSERT INTO subject_subject_types VALUES (4, 1);
INSERT INTO subject_subject_types VALUES (4, 3);

-- Operating Systems: Lecture + Laboratory
INSERT INTO subject_subject_types VALUES (5, 1);
INSERT INTO subject_subject_types VALUES (5, 3);

-- English: Lecture + Exercise
INSERT INTO subject_subject_types VALUES (6, 1);
INSERT INTO subject_subject_types VALUES (6, 2);
```

Fifteen students:
```
INSERT INTO students VALUES (1, 'Karla', 'Budimir');
INSERT INTO students VALUES (2, 'Ana', 'Horvat');
INSERT INTO students VALUES (3, 'Ivan', 'Kovac');
INSERT INTO students VALUES (4, 'Marta', 'Novak');
INSERT INTO students VALUES (5, 'Luka', 'Maric');
INSERT INTO students VALUES (6, 'Petra', 'Babic');
```

```
INSERT INTO students VALUES (7, 'Marko', 'Peric');
INSERT INTO students VALUES (8, 'Lucija', 'Tomic');
INSERT INTO students VALUES (9, 'Filip', 'Juric');
INSERT INTO students VALUES (10, 'Sara', 'Knezevic');
INSERT INTO students VALUES (11, 'Josip', 'Bosnic');
INSERT INTO students VALUES (12, 'Veronika', 'Budimir');
INSERT INTO students VALUES (13, 'Nina', 'Vidovic');
INSERT INTO students VALUES (14, 'David', 'Simic');
INSERT INTO students VALUES (15, 'Ema', 'Pavic');
```

Four teachers:

```
INSERT INTO teachers VALUES (1, 'Marina', 'Ivic');
INSERT INTO teachers VALUES (2, 'Tomislav', 'Basic');
INSERT INTO teachers VALUES (3, 'Ivana', 'Kralj');
INSERT INTO teachers VALUES (4, 'Petar', 'Matic');
```

Teacher' s salary and employment type:

```
INSERT INTO teacher_salary VALUES (1, 1800, 'Full-time');
INSERT INTO teacher_salary VALUES (2, 1500, 'Part-time');
INSERT INTO teacher_salary VALUES (3, 2000, 'Full-time');
INSERT INTO teacher_salary VALUES (4, 1200, 'Contract');
```

What subjects each teacher teaches:

```
INSERT INTO teacher_subjects VALUES (1, 1);
INSERT INTO teacher_subjects VALUES (1, 5);
INSERT INTO teacher_subjects VALUES (2, 2);
INSERT INTO teacher_subjects VALUES (3, 3);
INSERT INTO teacher_subjects VALUES (3, 6);
INSERT INTO teacher_subjects VALUES (4, 4);
```

Inserting grades:

```
-- grade_id, student_id, subject_id, type_id, grade

INSERT INTO grades VALUES (1, 1, 1, 1, 5);
INSERT INTO grades VALUES (2, 1, 1, 3, 5);
INSERT INTO grades VALUES (3, 1, 2, 1, 4);
INSERT INTO grades VALUES (4, 1, 2, 2, 5);

INSERT INTO grades VALUES (5, 2, 1, 1, 3);
INSERT INTO grades VALUES (6, 2, 1, 3, 4);
INSERT INTO grades VALUES (7, 2, 3, 1, 4);
INSERT INTO grades VALUES (8, 2, 3, 3, 5);

INSERT INTO grades VALUES (9, 3, 2, 1, 5);
INSERT INTO grades VALUES (10, 3, 2, 2, 4);
INSERT INTO grades VALUES (11, 3, 4, 1, 3);
INSERT INTO grades VALUES (12, 3, 4, 3, 4);
```

```
INSERT INTO grades VALUES (13, 4, 3, 1, 3);
INSERT INTO grades VALUES (14, 4, 3, 3, 4);
INSERT INTO grades VALUES (15, 4, 6, 1, 4);
INSERT INTO grades VALUES (16, 4, 6, 2, 5);

INSERT INTO grades VALUES (17, 5, 1, 1, 2);
INSERT INTO grades VALUES (18, 5, 1, 3, 3);
INSERT INTO grades VALUES (19, 5, 5, 1, 3);
INSERT INTO grades VALUES (20, 5, 5, 3, 4);

INSERT INTO grades VALUES (21, 6, 4, 1, 4);
INSERT INTO grades VALUES (22, 6, 4, 3, 5);
INSERT INTO grades VALUES (23, 6, 5, 1, 3);
INSERT INTO grades VALUES (24, 6, 5, 3, 4);

INSERT INTO grades VALUES (25, 7, 6, 1, 5);
INSERT INTO grades VALUES (26, 7, 6, 2, 4);
INSERT INTO grades VALUES (27, 7, 2, 1, 3);
INSERT INTO grades VALUES (28, 7, 2, 2, 4);

INSERT INTO grades VALUES (29, 8, 3, 1, 5);
INSERT INTO grades VALUES (30, 8, 3, 3, 5);
INSERT INTO grades VALUES (31, 8, 5, 1, 4);
INSERT INTO grades VALUES (32, 8, 5, 3, 3);

INSERT INTO grades VALUES (33, 9, 1, 1, 4);
INSERT INTO grades VALUES (34, 9, 1, 3, 5);
INSERT INTO grades VALUES (35, 9, 6, 1, 3);
INSERT INTO grades VALUES (36, 9, 6, 2, 4);

INSERT INTO grades VALUES (37, 10, 4, 1, 5);
INSERT INTO grades VALUES (38, 10, 4, 3, 4);
INSERT INTO grades VALUES (39, 10, 2, 1, 5);
INSERT INTO grades VALUES (40, 10, 2, 2, 5);

INSERT INTO grades VALUES (41, 11, 1, 1, 4);
INSERT INTO grades VALUES (42, 11, 1, 3, 5);
INSERT INTO grades VALUES (43, 11, 2, 1, 3);
INSERT INTO grades VALUES (44, 11, 2, 2, 4);

INSERT INTO grades VALUES (45, 12, 3, 1, 5);
INSERT INTO grades VALUES (46, 12, 3, 3, 4);
INSERT INTO grades VALUES (47, 12, 6, 1, 5);
INSERT INTO grades VALUES (48, 12, 6, 2, 5);

INSERT INTO grades VALUES (49, 13, 4, 1, 3);
INSERT INTO grades VALUES (50, 13, 4, 3, 4);
```

```
INSERT INTO grades VALUES (51, 13, 5, 1, 3);
INSERT INTO grades VALUES (52, 13, 5, 3, 4);

INSERT INTO grades VALUES (53, 14, 1, 1, 5);
INSERT INTO grades VALUES (54, 14, 1, 3, 5);
INSERT INTO grades VALUES (55, 14, 3, 1, 4);
INSERT INTO grades VALUES (56, 14, 3, 3, 5);

INSERT INTO grades VALUES (57, 15, 2, 1, 4);
INSERT INTO grades VALUES (58, 15, 2, 2, 4);
INSERT INTO grades VALUES (59, 15, 6, 1, 5);
INSERT INTO grades VALUES (60, 15, 6, 2, 4);

COMMIT;
```

## 1. Displaying all data from each table

```
SELECT * FROM subject_types;
SELECT * FROM subjects;
SELECT * FROM subject_subject_types;
SELECT * FROM students;
SELECT * FROM teachers;
SELECT * FROM teacher_salary;
SELECT * FROM teacher_subjects;
SELECT * FROM grades;
```

The output:

```
TYPE_ID TYPE_NAME
---------- ----------------------------------------
         1 Lecture
         2 Exercise
         3 Laboratory

SUBJECT_ID SUBJECT_NAME
---------- --------------------------------------------------------------------------------
         1 Databases
         2 Mathematics
         3 Programming
         4 Computer Networks
         5 Operating Systems
         6 English

6 rows selected.
```

```
SUBJECT_ID   TYPE_ID
---------- ----------
         1          1
         1          3
         2          1
         2          2
         3          1
         3          3
         4          1
         4          3
         5          1
         5          3
         6          1

SUBJECT_ID   TYPE_ID
---------- ----------
         6          2

12 rows selected.


STUDENT_ID FIRST_NAME                               LAST_NAME
---------- ---------------------------------------- ----------------------------------------
         1 Karla                                    Budimir
         2 Ana                                      Horvat
         3 Ivan                                     Kovac
         4 Marta                                    Novak
         5 Luka                                     Maric
         6 Petra                                    Babic
         7 Marko                                    Peric
         8 Lucija                                   Tomic
         9 Filip                                    Juric
        10 Sara                                     Knezevic
        11 Josip                                    Bosnic

STUDENT_ID FIRST_NAME                               LAST_NAME
---------- ---------------------------------------- ----------------------------------------
        12 Veronika                                 Budimir
        13 Nina                                     Vidovic
        14 David                                    Simic
        15 Ema                                      Pavic

15 rows selected.
```

| TEACHER_ID | FIRST_NAME | LAST_NAME |
|---|---|---|
| 1 | Marina | Ivic |
| 2 | Tomislav | Basic |
| 3 | Ivana | Kralj |
| 4 | Petar | Matic |

| TEACHER_ID | SALARY | EMPLOYMENT_TYPE |
|---|---|---|
| 1 | 1800 | Full-time |
| 2 | 1500 | Part-time |
| 3 | 2000 | Full-time |
| 4 | 1200 | Contract |

| TEACHER_ID | SUBJECT_ID |
|---|---|
| 1 | 1 |
| 1 | 5 |
| 2 | 2 |
| 3 | 3 |
| 3 | 6 |
| 4 | 4 |

6 rows selected.

| GRADE_ID | STUDENT_ID | SUBJECT_ID | TYPE_ID | GRADE |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 5 |
| 2 | 1 | 1 | 3 | 5 |
| 3 | 1 | 2 | 1 | 4 |
| 4 | 1 | 2 | 2 | 5 |
| 5 | 2 | 1 | 1 | 3 |
| 6 | 2 | 1 | 3 | 4 |
| 7 | 2 | 3 | 1 | 4 |
| 8 | 2 | 3 | 3 | 5 |
| 9 | 3 | 2 | 1 | 5 |
| 10 | 3 | 2 | 2 | 4 |
| 11 | 3 | 4 | 1 | 3 |

| GRADE_ID | STUDENT_ID | SUBJECT_ID | TYPE_ID | GRADE |
|---|---|---|---|---|
| 12 | 3 | 4 | 3 | 4 |
| 13 | 4 | 3 | 1 | 3 |
| 14 | 4 | 3 | 3 | 4 |
| 15 | 4 | 6 | 1 | 4 |
| 16 | 4 | 6 | 2 | 5 |
| 17 | 5 | 1 | 1 | 2 |
| 18 | 5 | 1 | 3 | 3 |
| 19 | 5 | 5 | 1 | 3 |
| 20 | 5 | 5 | 3 | 4 |
| 21 | 6 | 4 | 1 | 4 |
| 22 | 6 | 4 | 3 | 5 |

| GRADE_ID | STUDENT_ID | SUBJECT_ID | TYPE_ID | GRADE |
|---|---|---|---|---|
| 23 | 6 | 5 | 1 | 3 |
| 24 | 6 | 5 | 3 | 4 |
| 25 | 7 | 6 | 1 | 5 |
| 26 | 7 | 6 | 2 | 4 |
| 27 | 7 | 2 | 1 | 3 |
| 28 | 7 | 2 | 2 | 4 |
| 29 | 8 | 3 | 1 | 5 |
| 30 | 8 | 3 | 3 | 5 |
| 31 | 8 | 5 | 1 | 4 |
| 32 | 8 | 5 | 3 | 3 |
| 33 | 9 | 1 | 1 | 4 |

| GRADE_ID | STUDENT_ID | SUBJECT_ID | TYPE_ID | GRADE |
|---|---|---|---|---|
| 34 | 9 | 1 | 3 | 5 |
| 35 | 9 | 6 | 1 | 3 |
| 36 | 9 | 6 | 2 | 4 |
| 37 | 10 | 4 | 1 | 5 |
| 38 | 10 | 4 | 3 | 4 |
| 39 | 10 | 2 | 1 | 5 |
| 40 | 10 | 2 | 2 | 5 |
| 41 | 11 | 1 | 1 | 4 |
| 42 | 11 | 1 | 3 | 5 |
| 43 | 11 | 2 | 1 | 3 |
| 44 | 11 | 2 | 2 | 4 |

| GRADE_ID | STUDENT_ID | SUBJECT_ID | TYPE_ID | GRADE |
|---|---|---|---|---|
| 45 | 12 | 3 | 1 | 5 |
| 46 | 12 | 3 | 3 | 4 |
| 47 | 12 | 6 | 1 | 5 |
| 48 | 12 | 6 | 2 | 5 |
| 49 | 13 | 4 | 1 | 3 |
| 50 | 13 | 4 | 3 | 4 |
| 51 | 13 | 5 | 1 | 3 |
| 52 | 13 | 5 | 3 | 4 |
| 53 | 14 | 1 | 1 | 5 |
| 54 | 14 | 1 | 3 | 5 |
| 55 | 14 | 3 | 1 | 4 |

| GRADE_ID | STUDENT_ID | SUBJECT_ID | TYPE_ID | GRADE |
|---|---|---|---|---|
| 56 | 14 | 3 | 3 | 5 |
| 57 | 15 | 2 | 1 | 4 |
| 58 | 15 | 2 | 2 | 4 |
| 59 | 15 | 6 | 1 | 5 |
| 60 | 15 | 6 | 2 | 4 |

60 rows selected.

## 2. Display the average grade from each subject

```
SELECT s.subject_name,
    ROUND(AVG(g.grade), 2) AS average_grade
FROM subjects s
JOIN grades g
  ON s.subject_id = g.subject_id
GROUP BY s.subject_name
ORDER BY s.subject_name;
```

The output:

| SUBJECT_NAME | AVERAGE_GRADE |
|---|---|
| Computer Networks | 4 |
| Databases | 4.17 |
| English | 4.4 |
| Mathematics | 4.17 |
| Operating Systems | 3.5 |
| Programming | 4.4 |

6 rows selected.

## 3. Display the average grade for each student

This query ignores the type (Lecture, Laboratory and Exerise) and calculates the average treating all grades the same:

```
SELECT st.student_id,
    st.first_name,
    st.last_name,
    ROUND(AVG(g.grade), 2) AS average_grade
FROM students st
JOIN grades g
  ON st.student_id = g.student_id
GROUP BY st.student_id,
      st.first_name,
      st.last_name
ORDER BY st.student_id;
```

The output:

| STUDENT_ID | FIRST_NAME | LAST_NAME | AVERAGE_GRADE |
|---|---|---|---|
| 1 | Karla | Budimir | 4.75 |
| 2 | Ana | Horvat | 4 |
| 3 | Ivan | Kovac | 4 |
| 4 | Marta | Novak | 4 |
| 5 | Luka | Maric | 3 |
| 6 | Petra | Babic | 4 |
| 7 | Marko | Peric | 4 |
| 8 | Lucija | Tomic | 4.25 |
| 9 | Filip | Juric | 4 |
| 10 | Sara | Knezevic | 4.75 |
| 11 | Josip | Bosnic | 4 |

| STUDENT_ID | FIRST_NAME | LAST_NAME | AVERAGE_GRADE |
|---|---|---|---|
| 12 | Veronika | Budimir | 4.75 |
| 13 | Nina | Vidovic | 3.5 |
| 14 | David | Simic | 4.75 |
| 15 | Ema | Pavic | 4.25 |

15 rows selected.

## 4. Display the average grade for one type of subject for each student

This query displays the average grade for subject English, but only Exercise type (only for students enrolled in that subject):

```
SELECT st.student_id,
    st.first_name,
    st.last_name,
    ROUND(AVG(g.grade), 2) AS average_grade
FROM students st
JOIN grades g
  ON st.student_id = g.student_id
JOIN subjects s
  ON g.subject_id = s.subject_id
JOIN subject_types t
  ON g.type_id = t.type_id
WHERE s.subject_name = 'English'
 AND t.type_name = 'Exercise'
GROUP BY st.student_id,
     st.first_name,
     st.last_name
ORDER BY st.student_id;
```

The output:

| STUDENT_ID | FIRST_NAME | LAST_NAME | AVERAGE_GRADE |
|---|---|---|---|
| 4 | Marta | Novak | 5 |
| 7 | Marko | Peric | 4 |
| 9 | Filip | Juric | 4 |
| 12 | Veronika | Budimir | 5 |
| 15 | Ema | Pavic | 4 |